```
!pip install diffusers transformers accelerate
```

```
Requirement already satisfied: diffusers in
/usr/local/lib/python3.12/dist-packages (0.37.1)
Requirement already satisfied: transformers in
/usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: accelerate in
/usr/local/lib/python3.12/dist-packages (1.13.0)
Requirement already satisfied: importlib-metadata in
/usr/local/lib/python3.12/dist-packages (from diffusers) (8.7.1)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from diffusers) (3.25.2)
Requirement already satisfied: httpx<1.0.0 in
/usr/local/lib/python3.12/dist-packages (from diffusers) (0.28.1)
Requirement already satisfied: huggingface-hub<2.0,>=0.34.0 in
/usr/local/lib/python3.12/dist-packages (from diffusers) (1.10.1)
Requirement already satisfied: numpy in
/usr/local/lib/python3.12/dist-packages (from diffusers) (2.0.2)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from diffusers) (2025.11.3)
Requirement already satisfied: requests in
/usr/local/lib/python3.12/dist-packages (from diffusers) (2.32.4)
Requirement already satisfied: safetensors>=0.3.1 in
/usr/local/lib/python3.12/dist-packages (from diffusers) (0.7.0)
Requirement already satisfied: Pillow in
/usr/local/lib/python3.12/dist-packages (from diffusers) (11.3.0)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: tqdm>=4.27 in
/usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: psutil in
/usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in
/usr/local/lib/python3.12/dist-packages (from accelerate)
(2.10.0+cu128)
Requirement already satisfied: anyio in
/usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->diffusers)
(4.13.0)
Requirement already satisfied: certifi in
/usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->diffusers)
(2026.2.25)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->diffusers)
```

```
(1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-
packages (from httpx<1.0.0->diffusers) (3.11)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1.0.0->diffusers) (0.16.0)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=0.34.0->diffusers) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=0.34.0->diffusers) (1.4.3)
Requirement already satisfied: typer in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=0.34.0->diffusers) (0.24.1)
Requirement already satisfied: typing-extensions>=4.1.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=0.34.0->diffusers) (4.15.0)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (3.1.6)
Requirement already satisfied: cuda-bindings==12.9.4 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.93)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.90)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.90)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in
```

```
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (11.3.3.83)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (11.7.3.90)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.5.8.93)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (3.4.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.93)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (1.13.1.3)
Requirement already satisfied: triton==3.6.0 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4-
>torch>=2.0.0->accelerate) (1.5.2)
Requirement already satisfied: zipp>=3.20 in
/usr/local/lib/python3.12/dist-packages (from importlib-metadata-
>diffusers) (3.23.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests->diffusers)
(3.4.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests->diffusers)
(2.5.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3-
>torch>=2.0.0->accelerate) (1.3.0)
Requirement already satisfied: click>=8.2.1 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
```

```
hub<2.0,>=0.34.0->diffusers) (8.3.2)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub<2.0,>=0.34.0->diffusers) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub<2.0,>=0.34.0->diffusers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub<2.0,>=0.34.0->diffusers) (0.0.4)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch>=2.0.0-
>accelerate) (3.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
>huggingface-hub<2.0,>=0.34.0->diffusers) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
>huggingface-hub<2.0,>=0.34.0->diffusers) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=12.3.0->typer->huggingface-hub<2.0,>=0.34.0->diffusers) (0.1.2)
```

```python
import torch
from diffusers import StableDiffusionPipeline

# Definimos el modelo v1.5 que es el estándar para educación
model_id = "runwayml/stable-diffusion-v1-5"

# Cargamos el pipeline completo
# Usamos float16 para que ocupe menos memoria y corra más rápido en la
T4
pipe = StableDiffusionPipeline.from_pretrained(
    model_id,
    torch_dtype=torch.float16
)

# Movemos el modelo a la GPU (CUDA)
pipe = pipe.to("cuda")

print("□ Modelo cargado y listo para usar en GPU")
```

```
Flax classes are deprecated and will be removed in Diffusers v1.0.0.
We recommend migrating to PyTorch classes or pinning your version of
Diffusers.
Flax classes are deprecated and will be removed in Diffusers v1.0.0.
We recommend migrating to PyTorch classes or pinning your version of
Diffusers.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py
```

```
:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
  warnings.warn(
```

{"model_id":"bd5154aaf76a4e4e95944e08622af1c2","version_major":2,"version_minor":0}

{"model_id":"a7aa895b08d040e593be2f8c42de7946","version_major":2,"version_minor":0}

{"model_id":"563a08abe5464b26b0c9f676283e8555","version_major":2,"version_minor":0}

```
Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.
```

{"model_id":"05db29f0883d4455802be13760d63ea6","version_major":2,"version_minor":0}

{"model_id":"765392af14ff4360876cdc46e2b317d7","version_major":2,"version_minor":0}

```
CLIPTextModel LOAD REPORT from: /root/.cache/huggingface/hub/models--
runwayml--stable-diffusion-v1-5/snapshots/
451f4fe16113bff5a5d2269ed5ad43b0592e9a14/text_encoder
Key                                | Status     |  |
-----------------------------------+------------+--+-
text_model.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
```

{"model_id":"fae2cc4cb0cd45718e754648d408a66c","version_major":2,"version_minor":0}

```
StableDiffusionSafetyChecker LOAD REPORT from:
/root/.cache/huggingface/hub/models--runwayml--stable-diffusion-v1-5/
snapshots/451f4fe16113bff5a5d2269ed5ad43b0592e9a14/safety_checker
Key                                                  | Status     |  |
-----------------------------------------------------+------------+--+-
```

```
vision_model.vision_model.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
```

```
 Modelo cargado y listo para usar en GPU
```

```
# --- PANEL DE CONTROL PARA LA ACTIVIDAD ---
prompt = "A futuristic cup of coffee, neon glowing liquid, highly
detailed"
pasos = 25          # EXPERIMENTO A: Cambiar a 1, 10, 25, 50 (U-Net)
obediencia = 7.5    # EXPERIMENTO B: Cambiar a 1, 7.5, 30 (CLIP)
semilla = 42        # EXPERIMENTO C: Cambiar para navegar el Espacio
Latente

# Configuramos la semilla para tener control total
generator = torch.Generator("cuda").manual_seed(semilla)

# Proceso de generación
# Aquí la U-Net empieza a quitar ruido basándose en el prompt
print(f"Generando imagen con {pasos} pasos y escala {obediencia}...")

image = pipe(
    prompt,
    num_inference_steps=pasos,
    guidance_scale=obediencia,
    generator=generator
).images[0]

# Guardar y mostrar
image.save("resultado_lab.png")
display(image)
```

```
Generando imagen con 25 pasos y escala 7.5...
```

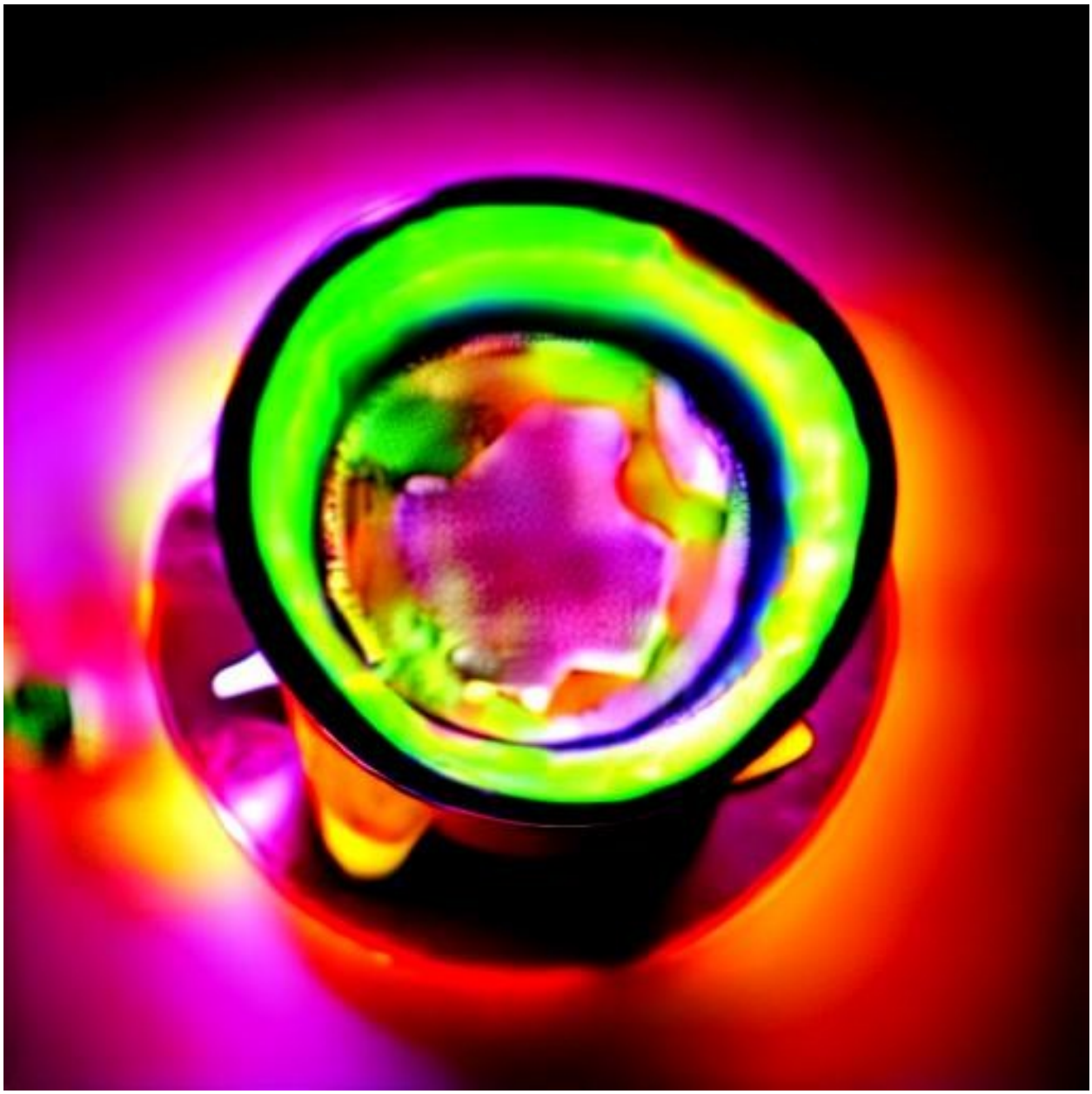
{"model_id":"26227049bf674e58a3e273b17739e756","version_major":2,"version_minor":0}

```
# --- PANEL DE CONTROL PARA LA ACTIVIDAD ---
prompt = "A futuristic cup of coffee, neon glowing liquid, highly
detailed"
pasos = 50         # EXPERIMENTO A: Cambiar a 1, 10, 25, 50 (U-Net)
obediencia = 89.0    # EXPERIMENTO B: Cambiar a 1, 7.5, 30 (CLIP)
semilla = 42        # EXPERIMENTO C: Cambiar para navegar el Espacio
Latente

# Configuramos la semilla para tener control total
generator = torch.Generator("cuda").manual_seed(semilla)
```

```python
# Proceso de generación
# Aquí la U-Net empieza a quitar ruido basándose en el prompt
print(f"Generando imagen con {pasos} pasos y escala {obediencia}...")

image = pipe(
    prompt,
    num_inference_steps=pasos,
    guidance_scale=obediencia,
    generator=generator
).images[0]

# Guardar y mostrar
image.save("resultado_lab.png")
display(image)
```

```
Generando imagen con 50 pasos y escala 89.0...
```

{"model_id":"2700e2e82abd4c5eb96ffe1aaeec91be","version_major":2,"version_minor":0}

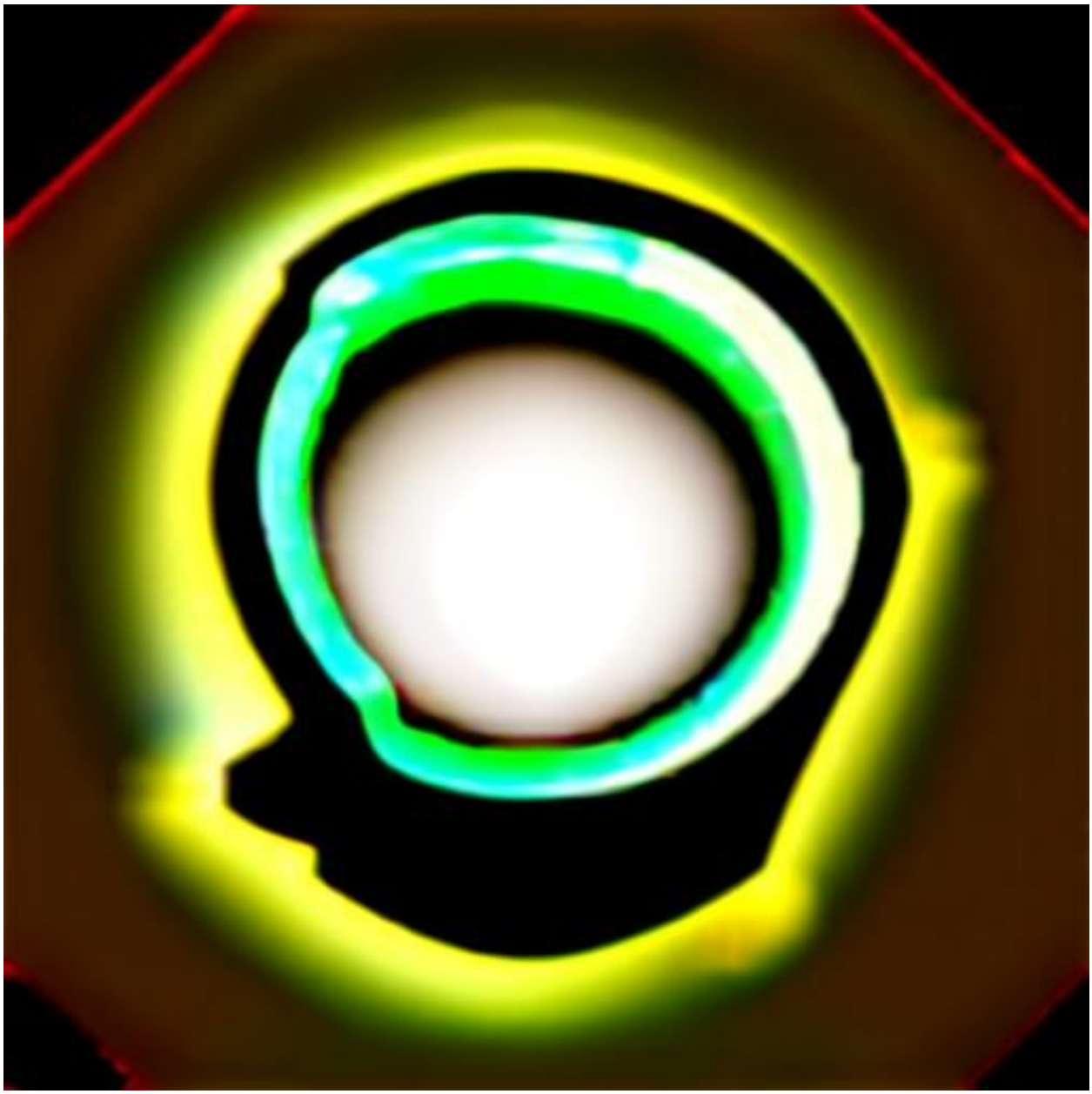


```
# --- PANEL DE CONTROL PARA LA ACTIVIDAD ---
prompt = "A futuristic cup of coffee, neon glowing liquid, highly
detailed"
pasos = 89         # EXPERIMENTO A: Cambiar a 1, 10, 25, 50 (U-Net)
obediencia = 98.0    # EXPERIMENTO B: Cambiar a 1, 7.5, 30 (CLIP)
semilla = 42        # EXPERIMENTO C: Cambiar para navegar el Espacio
Latente

# Configuramos la semilla para tener control total
generator = torch.Generator("cuda").manual_seed(semilla)
```

```python
# Proceso de generación
# Aquí la U-Net empieza a quitar ruido basándose en el prompt
print(f"Generando imagen con {pasos} pasos y escala {obediencia}...")

image = pipe(
    prompt,
    num_inference_steps=pasos,
    guidance_scale=obediencia,
    generator=generator
).images[0]

# Guardar y mostrar
image.save("resultado_lab.png")
display(image)
```

```
Generando imagen con 89 pasos y escala 98.0...
```

{"model_id":"d4811b9a2b5f4b729d41a3610ca340ef","version_major":2,"version_minor":0}



```
# --- PANEL DE CONTROL PARA LA ACTIVIDAD ---
prompt = "A futuristic cup of coffee, neon glowing liquid, highly
detailed"
pasos = 19          # EXPERIMENTO A: Cambiar a 1, 10, 25, 50 (U-Net)
obediencia = 1.75    # EXPERIMENTO B: Cambiar a 1, 7.5, 30 (CLIP)
semilla = 25        # EXPERIMENTO C: Cambiar para navegar el Espacio
Latente

# Configuramos la semilla para tener control total
generator = torch.Generator("cuda").manual_seed(semilla)

# Proceso de generación
```

```python
# Aquí la U-Net empieza a quitar ruido basándose en el prompt
print(f"Generando imagen con {pasos} pasos y escala {obediencia}...")

image = pipe(
    prompt,
    num_inference_steps=pasos,
    guidance_scale=obediencia,
    generator=generator
).images[0]

# Guardar y mostrar
image.save("resultado_lab.png")
display(image)
```

```
Generando imagen con 19 pasos y escala 1.75...
```

{"model_id":"756204e966e94f7da6c155c74de39739","version_major":2,"version_minor":0}

```
# --- PANEL DE CONTROL PARA LA ACTIVIDAD ---
prompt = "A futuristic cup of coffee, neon glowing liquid, highly
detailed"
pasos = 26        # EXPERIMENTO A: Cambiar a 1, 10, 25, 50 (U-Net)
obediencia = 34.0    # EXPERIMENTO B: Cambiar a 1, 7.5, 30 (CLIP)
semilla = 25        # EXPERIMENTO C: Cambiar para navegar el Espacio
Latente

# Configuramos la semilla para tener control total
generator = torch.Generator("cuda").manual_seed(semilla)

# Proceso de generación
```

```python
# Aquí la U-Net empieza a quitar ruido basándose en el prompt
print(f"Generando imagen con {pasos} pasos y escala {obediencia}...")

image = pipe(
    prompt,
    num_inference_steps=pasos,
    guidance_scale=obediencia,
    generator=generator
).images[0]

# Guardar y mostrar
image.save("resultado_lab.png")
display(image)
```

```
Generando imagen con 26 pasos y escala 34.0...
```

{"model_id":"11f9d760b3df42ba83e7c1bff58ab043","version_major":2,"version_minor":0}

```
# --- PANEL DE CONTROL PARA LA ACTIVIDAD ---
prompt = "A futuristic cup of coffee, neon glowing liquid, highly
detailed"
pasos = 28        # EXPERIMENTO A: Cambiar a 1, 10, 25, 50 (U-Net)
obediencia = 5.0    # EXPERIMENTO B: Cambiar a 1, 7.5, 30 (CLIP)
semilla = 18       # EXPERIMENTO C: Cambiar para navegar el Espacio
Latente

# Configuramos la semilla para tener control total
generator = torch.Generator("cuda").manual_seed(semilla)

# Proceso de generación
```

```python
# Aquí la U-Net empieza a quitar ruido basándose en el prompt
print(f"Generando imagen con {pasos} pasos y escala {obediencia}...")

image = pipe(
    prompt,
    num_inference_steps=pasos,
    guidance_scale=obediencia,
    generator=generator
).images[0]

# Guardar y mostrar
image.save("resultado_lab.png")
display(image)
```

```
Generando imagen con 28 pasos y escala 5.0...
```

{"model_id":"19edae3d807e42f1887f7c3d9851a39d","version_major":2,"version_minor":0}

```
# --- PANEL DE CONTROL PARA LA ACTIVIDAD ---
prompt = "A wite young woman"
pasos = 25          # EXPERIMENTO A: Cambiar a 1, 10, 25, 50 (U-Net)
obediencia = 7.5    # EXPERIMENTO B: Cambiar a 1, 7.5, 30 (CLIP)
semilla = 42        # EXPERIMENTO C: Cambiar para navegar el Espacio
Latente

# Configuramos la semilla para tener control total
generator = torch.Generator("cuda").manual_seed(semilla)

# Proceso de generación
# Aquí la U-Net empieza a quitar ruido basándose en el prompt
```

```python
print(f"Generando imagen con {pasos} pasos y escala {obediencia}...")

image = pipe(
    prompt,
    num_inference_steps=pasos,
    guidance_scale=obediencia,
    generator=generator
).images[0]

# Guardar y mostrar
image.save("resultado_lab.png")
display(image)
```

```
Generando imagen con 25 pasos y escala 7.5...
```

{"model_id":"a642cd6cf416451fb2d000587ca21a17","version_major":2,"version_minor":0}

```
# --- PROMPT 1: EL GRITO (EXPERIMENTO DE OBEDIENCIA) ---
prompt = "A hyper-realistic astronaut eating an apple on the moon, oil
painting style."
pasos = 25
semilla = 42
lista_obediencia = [1.0, 7.5, 22.5, 30.0] # Valores a probar

for obediencia in lista_obediencia:
    print(f"Generando imagen | Pasos: {pasos} | Semilla: {semilla} |
Obediencia: {obediencia}...")

    # Reiniciamos el generador en cada ciclo para que la semilla sea
```

```
exactamente la misma
    generator = torch.Generator("cuda").manual_seed(semilla)

    image = pipe(
        prompt,
        num_inference_steps=pasos,
        guidance_scale=obediencia,
        generator=generator
    ).images[0]

    # Guardar y mostrar
    nombre_archivo = f"resultado_p1_obediencia_{obediencia}.png"
    image.save(nombre_archivo)
    print(f"Guardado como: {nombre_archivo}")
    display(image)
```

```
Generando imagen | Pasos: 25 | Semilla: 42 | Obediencia: 1.0...
```

{"model_id":"d8122d6ed8724210a4973f59ce58bd91","version_major":2,"version_minor":0}

```
Guardado como: resultado_p1_obediencia_1.0.png
```

```
Generando imagen | Pasos: 25 | Semilla: 42 | Obediencia: 7.5...
```


```
{"model_id":"80c7178c963744de9cf78bf457b6a480","version_major":2,"version_minor":0}
```


```
Guardado como: resultado_p1_obediencia_7.5.png
```

```
Generando imagen | Pasos: 25 | Semilla: 42 | Obediencia: 22.5...

{"model_id":"2675549ea14d460daba00d541ed2a88a","version_major":2,"version_minor":0}

Guardado como: resultado_p1_obediencia_22.5.png
```

```
Generando imagen | Pasos: 25 | Semilla: 42 | Obediencia: 30.0...
```


```
{"model_id":"d13d010e9b6f4be3b04b614594950dbf","version_major":2,"version_minor":0}
```


```
Guardado como: resultado_p1_obediencia_30.0.png
```

```
# --- PROMPT 2: MEZCLA DE CONCEPTOS (NUEVA SEMILLA) ---
prompt = "A hybrid between a vintage typewriter and a lush green
forest, moss growing on keys, hyper-detailed"
pasos = 25
semilla = 555       # Modificada a un valor entre 120 y 999
lista_obediencia = [1.0, 7.5, 10.0]

print(f"--- INICIANDO CON NUEVA SEMILLA: {semilla} ---")

for obediencia in lista_obediencia:
    print(f"Generando imagen | Pasos: {pasos} | Semilla: {semilla} |
Obediencia: {obediencia}...")
```

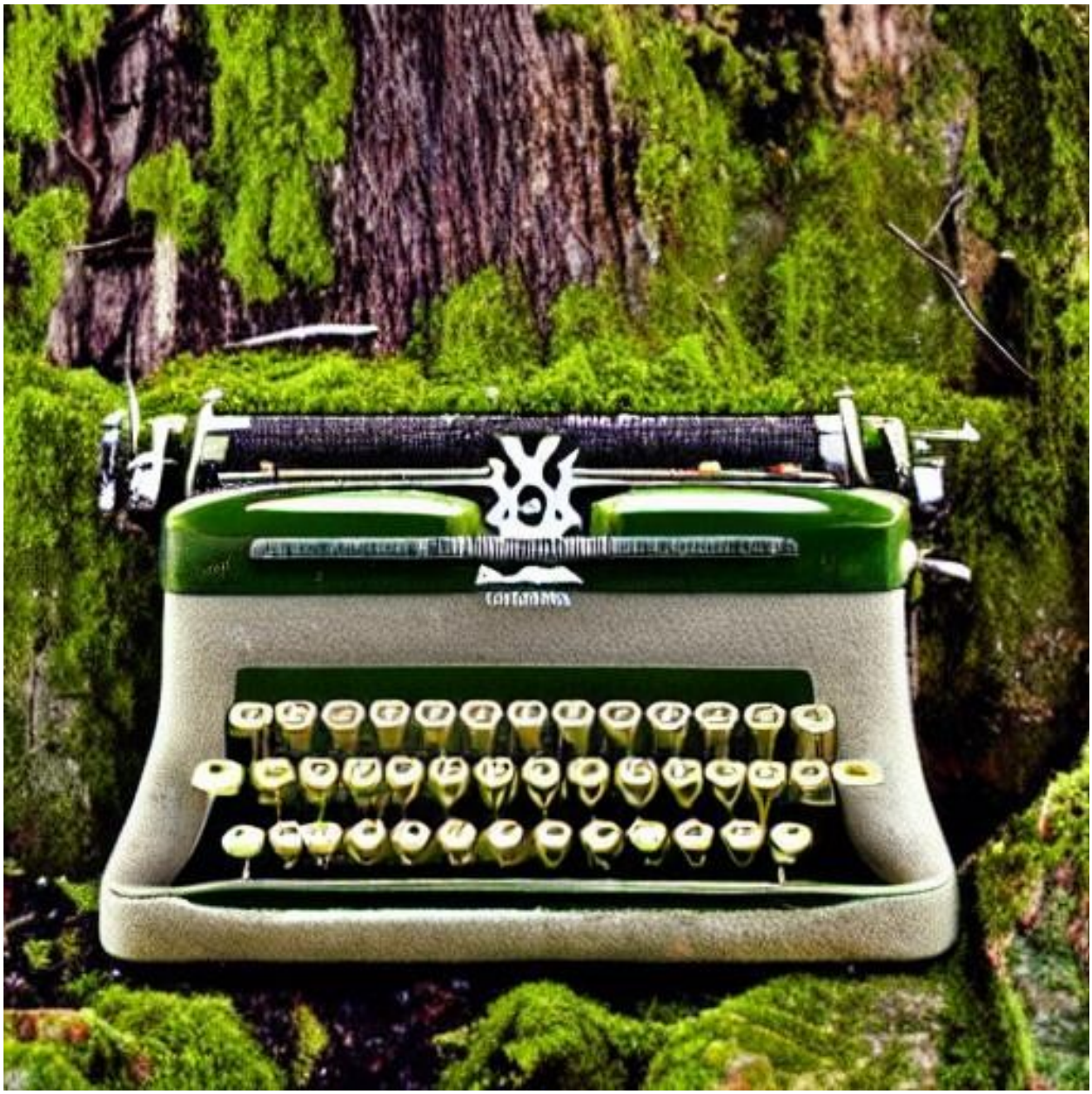
```
    generator = torch.Generator("cuda").manual_seed(semilla)

    image = pipe(
        prompt,
        num_inference_steps=pasos,
        guidance_scale=obediencia,
        generator=generator
    ).images[0]

    nombre_archivo = f"resultado_p2_obediencia_{obediencia}.png"
    image.save(nombre_archivo)
    display(image)
```

```
--- INICIANDO CON NUEVA SEMILLA: 555 ---
Generando imagen | Pasos: 25 | Semilla: 555 | Obediencia: 1.0...
```

{"model_id":"c7272b4889854e2f9329a45175b964f4","version_major":2,"version_minor":0}

```
Generando imagen | Pasos: 25 | Semilla: 555 | Obediencia: 7.5...
```

{"model_id":"fe0f6e171dd04f2890ba003ad1160ba8","version_major":2,"version_minor":0}



```
Generando imagen | Pasos: 25 | Semilla: 555 | Obediencia: 10.0...
```


```
{"model_id":"fea11851015b4a3e960973f7c2f2c10c","version_major":2,"version_minor":0}
```

```
# --- PROMPT 3: PROMPT PERSONALIZADO EN INGLÉS ---
prompt = "A giant pepperoni pizza floating in a cyberpunk city
skyline, neon lights reflecting on the melting cheese, cinematic
angle, hyper-detailed"
pasos = 25
obediencia = 7.5
semilla = 42

print(f"Generando imagen base | Pasos: {pasos} | Semilla: {semilla} |
Obediencia: {obediencia}...")

generator = torch.Generator("cuda").manual_seed(semilla)
```

```
image = pipe(
    prompt,
    num_inference_steps=pasos,
    guidance_scale=obediencia,
    generator=generator
).images[0]

image.save("resultado_p3_pizza_cyberpunk.png")
display(image)
```

```
Generando imagen base | Pasos: 25 | Semilla: 42 | Obediencia: 7.5...
```

{"model_id":"5d5c0b0bef6a4bab885b38178a4918e7","version_major":2,"version_minor":0}

```
# --- PROMPT 4: EXPERIMENTO DE RUPTURA DE OBEDIENCIA ---
prompt = "A plate of spicy buffalo chicken wings surrounded by epic
fire and freezing ice elements, cinematic lighting, 8k resolution"
pasos = 25
semilla = 42
lista_obediencia = [7.5, 20.0] # Contrastando obediencia normal vs
extrema

for obediencia in lista_obediencia:
    print(f"Generando contraste | Pasos: {pasos} | Semilla: {semilla}
| Obediencia: {obediencia}...")
```

```
    generator = torch.Generator("cuda").manual_seed(semilla)

    image = pipe(
        prompt,
        num_inference_steps=pasos,
        guidance_scale=obediencia,
        generator=generator
    ).images[0]

    nombre_archivo =
f"resultado_p4_alitas_obediencia_{obediencia}.png"
    image.save(nombre_archivo)
    display(image)
```

```
Generando contraste | Pasos: 25 | Semilla: 42 | Obediencia: 7.5...
```

{"model_id":"8ae7079dce9e426fad7586a241453493","version_major":2,"version_minor":0}

```
Generando contraste | Pasos: 25 | Semilla: 42 | Obediencia: 20.0...
```

{"model_id":"0a259643c7ec43deaabdb740c5f58cd8","version_major":2,"version_minor":0}